Type MS · Turning sin and cos equations into tan

Answer 1 of 9 · Question 2 part whole · 5 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

Question	Scheme	Mark
2	$\sin(2\theta-20)^\circ = \sqrt{3}\cos(2\theta-20)^\circ \Rightarrow \tan(2\theta-20)^\circ = \sqrt{3}$	M1
	$(2\theta-20)^\circ = 60^\circ,\ 240^\circ,\ 420^\circ,...$	M1A1
	$\theta = \frac{'60'+20}{2} = 40^\circ,\quad \theta = \frac{'240'+20}{2} = 130^\circ$	M1A1 [5]
	Total	**5 marks**

Mark	Notes
M1	For using the identity $\frac{\sin A}{\cos A} = \tan A$ to reach $\tan(2\theta-20)^\circ = k$ where k is a numerical value $\sin(2\theta-20)^\circ = \sqrt{3}\cos(2\theta-20)^\circ \Rightarrow \tan(2\theta-20)^\circ = \sqrt{3}$
M1	For finding at least one **correct** angle for $(2\theta-20)^\circ$ $(2\theta-20)^\circ = 60^\circ,\ 240^\circ,\ 420^\circ,...$ Allow even for eg., -120° This mark can be implied by correct final answers.
A1	For both of the angles 60° and 240° Ignore any extra values, even if within range. $0 \leqslant (2\theta-20) \leqslant 360$ This mark can be implied by correct final answers.
M1	For correct processing of their values for $(2\theta-20)^\circ$ $\theta = \frac{'60'+20}{2} = ...$ **or** $\theta = \frac{'240'+20}{2} = ...$
A1 [5]	For both correct values of $\theta =$ 40 and 130 Ignore other angles out of range, penalise extra angles within the range by the loss of this A mark

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Answer 1 of 9 · Question 2 part whole · 5 marks

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Question	Scheme	Mark

Type MS · Turning sin and cos equations into tan

Answer 2 of 9 · Question 10 part b · 3 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Question number		
10 (a)	$(x+\frac{\pi}{3})=\frac{\pi}{3}$ or $\frac{2\pi}{3}$ or $\frac{7\pi}{3}$ $(x+\frac{\pi}{3})=\frac{\pi}{3}$ and $\frac{2\pi}{3}$ and $\frac{7\pi}{3}$ $x=0,\frac{\pi}{3},2\pi$	M1 A1 A1 (3)
(b)	$\tan\theta=-\frac{5}{3}$ $\theta = -59^\circ, -239^\circ, 121^\circ, 301^\circ$	M1 M1 A1 (3)
(c)	$1+\sin 2y-2(1-\sin^2 2y)=0$ $2\sin^2 2y+\sin 2y-1=0$ $(\sin 2y+1)(2\sin 2y-1)=0$ $\sin 2y=-1$ or $\sin 2y=\frac{1}{2}$ $2y=-90^\circ, (30^\circ), (150^\circ), -330^\circ, -210^\circ$ $y=-45^\circ, -105^\circ, -165^\circ$	M1 A1 dM1 A1 A1 (5) **[11]**

Part	Mark	Additional Guidance
(a)	M1	Any one of the three indicated angles, radians only, ignore any other angles.
	A1	For all three indicated angles, ignore other angles out of the range $\frac{\pi}{3}\le x+\frac{\pi}{3}\le\frac{7\pi}{3}$
	A1	For all three angles, ignore angles out of range, A0 if additional angles in range.
(b)	M1	For $\tan\theta=k$. $k\neq0$, $k\neq\pm1$
	M1	Any one correct value, does not need to be to the nearest degree. Allow one correct value to imply the first M1. Ignore any other angles.
	A1	For all four angles, ignore angles out of range, A0 if additional angles in range. All four angles must be given to the nearest degree.
(c)	M1	For the correct use of $1-\sin^2 2y$ in the equation on the left or right side, equation doesn't need to be = 0.
	A1	Correct 3TQ, must be = 0 and a valid attempt to solve leading to sin $2y$ =
	dM1	$\sin 2y=-1$ or $\sin 2y=\frac{1}{2}$ (allow sin2y = a and b from a valid attempt to solve their 3TQ). Allow sin2y to be x or any other variable.
	A1	For minimum of 3 of the 5 values shown, (including the ones in brackets), ignore other angles outside the range -360° ≤ $2y$ ≤ 360°. Allow sight of 270 if -90° present.
	A1	For all 3 values shown. Ignore extras out of range. **Rounding answers (where accuracy is specified in the question)** Penalise only once per question for failing to round as instructed - ie giving more digits in the answers.

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Answer 2 of 9 · Question 10 part b · 3 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Turning sin and cos equations into tan

Answer 3 of 9 · Question 10 part a · 3 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

Number		
10 (a)	$\sin x + 1 = \cos x + 1 \Rightarrow \tan x = 1$	M1
	$x = \frac{\pi}{4}, \frac{5\pi}{4}$	A1 A1 (3)
(b)	Area R_1 :	
	$\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x + 1) - (\cos x + 1)\,dx = \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x - \cos x)\,dx$	M1
	$\left[-\cos x - \sin x\right]_{\frac{\pi}{4}}^{\frac{5\pi}{4}}$	M1
	$\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\right) - \left(-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\right) = 2\sqrt{2}$	M1 A1
	Area R_2 :	
	$\int_{\pi}^{\frac{5\pi}{4}} (\cos x + 1)\,dx + \int_{\frac{5\pi}{4}}^{\frac{3\pi}{2}} (\sin x + 1)\,dx$	M1
	$\left[\sin x + x\right]_{\pi}^{\frac{5\pi}{4}} + \left[-\cos x + x\right]_{\frac{5\pi}{4}}^{\frac{3\pi}{2}}$	A1
	$\left[\left(-\frac{\sqrt{2}}{2} + \frac{5\pi}{4}\right) - \pi\right] + \left[\frac{3\pi}{2} - \left(\frac{\sqrt{2}}{2} + \frac{5\pi}{4}\right)\right] = -\sqrt{2} + \frac{1}{2}\pi$	M1 A1
	area of R_1 : area of $R_2 = 2 : \left(\frac{\pi\sqrt{2}}{4} - 1\right)$ oe	A1 (9)
		Total 12 marks

Type MS · Turning sin and cos equations into tan

Answer 3 of 9 · Question 10 part a · 3 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For $\tan x = 1$
	A1	For $x = \frac{\pi}{4}$ [Allow 45°]
	A1	For $x = \frac{5\pi}{4}$ [Allow 225°]
(b)	M1	For stating $A = \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x + 1) - (\cos x + 1)\,dx$
	M1	For attempting to integrate their expression for the area. This must be correct for this mark. $\int (\sin x - \cos x)\,dx = -\cos x - \sin x$
	M1	For substitution of correct limits
	A1	For $2\sqrt{2}$
	M1	For attempting $\int f(x)dx + \int g(x)dx$ with the correct limits. $\int_{\pi}^{\frac{5\pi}{4}} (\cos x + 1)\,dx + \int_{\frac{5\pi}{4}}^{\frac{3\pi}{2}} (\sin x + 1)\,dx$
	M1	For attempting to integrate, which must be correct for this mark.
	M1	For substitution of their limits correctly.
	A1	For $-\sqrt{2} + \frac{1}{2}\pi$
	A1	For the simplified ratio $2 : \left\{\frac{\pi\sqrt{2}}{4} - 1\right\}$

Type MS · Turning sin and cos equations into tan

Answer 3 of 9 · Question 10 part a · 3 marks

2022 JAN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Turning sin and cos equations into tan

Answer 4 of 9 · Question 12 part b · 2 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Question number		
12 (a)	$(2\cos x = 0)$ $\quad (x =)\frac{\pi}{2}$ or 90 °	B1 (1)
(b)	$(2\cos x = 2\sin x)$ $\quad \tan x = 1$	M1
	$x = \frac{\pi}{4}$ or 45°	A1 (2)
(c)	$\int_{(0)}^{(\frac{\pi}{4})} (2\sin x)dx + \int_{(\frac{\pi}{4})}^{(\frac{\pi}{2})} (2\cos x)dx$	M1
	$\left[-2\cos x\right]_0^{\frac{\pi}{4}} + \left[2\sin x\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$	A1
	$\left(-\sqrt{2}+2\right)+\left(2-\sqrt{2}\right) = 4-2\sqrt{2}$	dM1
	$= 4-\sqrt{8}$ cao	A1cao cso (4) [7]

Part	Mark	Additional Guidance
(a)	B1	For $\frac{\pi}{2}$ or 90 degrees. Can also be shown as a coordinate – ignore any incorrect y coordinate.
(b)	M1	For $\tan x = 1$
	A1	For $\frac{\pi}{4}$ or 45 degrees. Can also be shown as a coordinate – ignore any incorrect y coordinate.
(c)	M1	For both integrals correctly shown, with an addition sign between them. Limits need not be shown. Must be 2 integrals shown. Can't be shown as one integral with incorrect limits.
	A1	For both functions correctly integrated. Limits need not be shown.
	dM1	For their limits clearly and correctly substituted in or for the numerical expression(s) shown in the MS. If mark awarded for substitution, both integrated expressions must have both limits correctly substituted. 0 must be the lower limit on the first integral. Allow ft of their $\frac{\pi}{4}$ (must be the upper limit on the first integral and the lower limit on the second and can be in degrees) and their $\frac{\pi}{2}$ (must be the upper limit on the second integral and can be in degrees).
	A1	cao cso A0 if degrees used in part c
		Note, can also be completed as either integral doubled – symmetry. M1 – correct integral stated with multiply by 2 evident or implicit later. A1 correctly integrated dM1 – as for main scheme, the multiply by 2 must be clearly shown. A1 as main scheme

Type MS · Turning sin and cos equations into tan

Answer 4 of 9 · Question 12 part b · 2 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Type MS · Turning sin and cos equations into tan

Answer 5 of 9 · Question 9 part c · 4 marks

2023 JUN · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

Question	Scheme	Marks
9(a)	$\cos 2A = \cos^2 A - \sin^2 A = \cos^2 A - (1-\cos^2 A) = 2\cos^2 A - 1$ OR $\cos 2A = \cos^2 A - \sin^2 A = (1-\sin^2 A) - \sin^2 A = 1 - 2\sin^2 A$ (i) $\cos^2 A = \frac{\cos 2A + 1}{2}$, (ii) $\sin^2 A = \frac{1-\cos 2A}{2}$ *	M1M1 A1A1cso [4]
(b)	$(2\sin x - \cos x)(\sin x - 3\cos x)$ $= 2\sin^2 x - 7\sin x\cos x + 3\cos^2 x$ $= 2\left(\frac{1-\cos 2x}{2}\right), \ldots\ldots\ldots\ldots, +3\left(\frac{\cos 2x+1}{2}\right)$ $= \ldots\ldots\ldots\ldots - \frac{7\sin 2x}{2} + \ldots\ldots\ldots\ldots$ $= \frac{1}{2}(\cos 2x - 7\sin 2x + 5)$*	M1 M1, M1 M1 A1 cso [5]
(c)	$y = \frac{1}{2}(\cos 2x - 7\sin 2x + 5) \Rightarrow \frac{dy}{dx} = -\sin 2x - 7\cos 2x = 0$ $\Rightarrow -\sin 2x = 7\cos 2x \Rightarrow \tan 2x = -7$ $2x = -81.869, \quad 98.130, \quad 278.130$ $\Rightarrow x = 49, \quad 139$	M1 M1 A1 A1 [4]
		Total 13 marks

Part	Mark	Notes
NB: Parts (a) and (b) are proofs with given answers. You must check every line of their working.		
(a)	M1	Uses the formula for $\cos(A+B) = \cos A\cos B - \sin A\sin B$ correctly.
	M1	Uses the identity $\cos^2 A = 1 - \sin^2 A$ or $\sin^2 A = 1 - \cos^2 A$
(i)	A1 cso	For the correct identity for either $\sin^2 A$ or $\cos^2 A$ with no errors seen
(ii)	A1 cso	For the correct identity for $\sin^2 A$ and $\cos^2 A$ with no errors seen
SC	If a candidate does not use $\cos(A+B) = \cos A\cos B - \sin A\sin B$ to find the required identities for $\sin^2 A$ or $\cos^2 A$ but uses them to show $\sin^2 A = \frac{1-\cos 2A}{2}$ or $\cos^2 A = \frac{\cos 2A+1}{2}$ apply the following scheme.	
	First M mark – M0	
	2nd M1	Uses the identity $\cos^2 A = 1 - \sin^2 A$ or $\sin^2 A = 1 - \cos^2 A$ For example:

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Answer 5 of 9 · Question 9 part c · 4 marks

2023 JUN · Paper 01 · extract 2 of 4 · find part c in this mark scheme extract

		$\cos^2 A = \frac{\cos 2A + 1}{2} = 1 - \sin^2 A$ $\sin^2 A = 1 - \frac{\cos 2A + 1}{2} = \frac{2 - (\cos 2A + 1)}{2} = \frac{1 - \cos 2A}{2}$
(i)/(ii)	**A1 cso**	For the correct identity for $\sin^2 A$ or $\cos^2 A$ with no errors seen
	A0	Not available
(b)	**M1**	Multiplies out the brackets with no errors.
	M1	Substitutes at least one of the given identities for $\cos 2x$ into their expansion [either for $\sin^2 x$ or $\cos^2 x$]. This is not a dependent mark, allow for substitution into an erroneous expression.
	M1	Substitutes both of the given identities for $\cos 2x$ into their expansion [for both $\sin^2 x$ and $\cos^2 x$]. This is not a dependent mark, allow for substitution into an erroneous expression. **NB: This is a B mark in Epen**
	M1	Uses the identity $\sin 2A = 2\sin A \cos A$ seen applied explicitly **before** the final answer. Look for, $k \sin x \cos x \rightarrow \frac{k \sin 2x}{2}$
	A1 cso	For the correct identity with no errors seen.
	ALT – Works backwards – these steps must contain no errors (to maintain compatibility with the main method. Starting from $\frac{1}{2}(\cos 2x - 7\sin 2x + 5)$ award as follows:	
	M1	Uses: $\cos 2x = \cos^2 x - \sin^2 x$
	M1	Uses: $5 = 5\cos^2 x + 5\sin^2 x$
	M1	Uses: $-7\sin 2x = 14 \sin x \cos x$
	M1	Collects like terms and divides by 2 to give: $3\cos^2 x - 7\sin x \cos x + 2\sin^2 x$
	A1	Factorises their expression to give: $(2\sin x - \cos x)(\sin x - 3\cos x)$
(c)	**M1**	For an attempt to differentiate the given equation. A minimally acceptable is defined as follows: $\frac{dy}{dx} = \pm k \sin 2x \pm l \cos 2x$ where k and l are integers **with no other terms**.
	M1	Uses the identity $\tan 2x = \frac{\sin 2x}{\cos 2x}$ This is an independent mark, but it must be used correctly following correct division of $\cos 2x$ throughout.
	A1	Find at least one **correct** value for $2x$ e.g., –81.869, 98.130, 278.130 etc **NB: This is an M mark in Epen.** [Accept – 81, 98, 278]
	A1	For $x = 49$ and 139 rounded to the nearest whole number or better.

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Answer 5 of 9 · Question 9 part c · 4 marks

2023 JUN · Paper 01 · extract 3 of 4 · find part c in this mark scheme extract

	ALT - Differentiates using product rule	
	M1	$y=(2\sin x-\cos x)(\sin x-3\cos x)$ $\frac{dy}{dx}=(2\sin x-\cos x)(\cos x+3\sin x)+(\sin x-3\cos x)(2\cos x+\sin x)$ $\frac{dy}{dx}=2\sin x\cos x+6\sin^2 x-\cos^2 x-3\sin x\cos x+$ $+2\sin x\cos x+\sin^2 x-6\cos^2 x-3\sin x\cos x$ $\frac{dy}{dx}=-2\sin x\cos x+6\sin^2 x-7\cos^2 x$ **Allow** $\frac{dy}{dx}=\pm p\sin x\cos x\pm q\sin^2 x\pm r\cos^2 x$ **NB: This will not factorise**
	M1	Divides through by $\cos^2 x$ provided they have an expression of the form $\frac{dy}{dx}=\pm p\sin x\cos x\pm q\sin^2 x\pm r\cos^2 x$ This gives : $\frac{dy}{dx}=-2\sin x\cos x+6\sin^2 x-7\cos^2 x=0 \quad \rightarrow \div\cos^2 x$ $-7-2\tan x+7\tan^2 x=0$ **AND** solves to give a value for $\tan x=\frac{1\mp 5\sqrt{2}}{7}$ or allow $\tan x=\frac{1\pm 5\sqrt{2}}{7}$
	A1	At least one correct value for x of for example, awrt 49.1….awrt – 40.9
	A1	For both 49° and 139°

Type MS · Turning sin and cos equations into tan

Answer 5 of 9 · Question 9 part c · 4 marks

2023 JUN · Paper 01 · extract 4 of 4 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Turning sin and cos equations into tan

Answer 6 of 9 · Question 11 part b · 5 marks

2024 JUN · Paper 01 (regional R) · find part b in this mark scheme extract

Question	Scheme	Marks
11(a)	$f(\theta)=(2\cos\theta-\sin\theta)(2\sin\theta+\cos\theta)$ $=4\sin\theta\cos\theta+2\cos^2\theta-2\sin^2\theta-\sin\theta\cos\theta$ $=3\sin\theta\cos\theta+2(\cos^2\theta-\sin^2\theta)\Rightarrow \frac{3}{2}\sin 2\theta,+2\cos 2\theta$ $=\frac{3}{2}\sin 2\theta+2\cos 2\theta$ *	M1 M1A1 cso [3]
(b)	$\frac{3}{2}\sin 2\theta+2\cos 2\theta+2=0\Rightarrow 3\sin 2\theta+4\cos 2\theta+4=0$ $\Rightarrow 3\sin 2\theta+4(\cos^2\theta-\sin^2\theta)+4(\sin^2\theta+\cos^2\theta)=0$ $\Rightarrow 6\sin\theta\cos\theta+8\cos^2\theta=0$ $\Rightarrow \cos\theta(6\sin\theta+8\cos\theta)=0$ $\Rightarrow \cos\theta=0\Rightarrow\theta=\frac{\pi}{2}$ * $\left[\Rightarrow \tan\theta=-\frac{4}{3}\right]$	M1M1 M1 M1 A1 cso [5]
(c)	$\text{Area}=\int_0^{\frac{\pi}{2}}\left(\frac{3}{2}\sin 2\theta+2\cos 2\theta+2\right)d\theta=\left[-\frac{3}{4}\cos 2\theta+\frac{2\sin 2\theta}{2}+2\theta\right]_0^{\frac{\pi}{2}}$ $=\left(-\frac{3}{4}\cos 2\left(\frac{\pi}{2}\right)+\frac{2\sin 2\left(\frac{\pi}{2}\right)}{2}+2\left(\frac{\pi}{2}\right)\right)-\left(-\frac{3}{4}\cos 0+\frac{2\sin 0}{2}+2\times 0\right)$ $=\left(\frac{3}{4}+\pi\right)-\left(-\frac{3}{4}\right)=\frac{3}{2}+\pi$	M1 M1 A1 [3]
		Total 11 marks

Type MS · Turning sin and cos equations into tan

Answer 7 of 9 · Question 8 part b.ii · 4 marks

2024 NOV · Paper 01 · find part b.ii in this mark scheme extract

Number		
8 (i) (a)	$\left(\tan 2A = \tan(A+A) =\right)\dfrac{\tan A+\tan A}{1-\tan A\tan A}$ $=\dfrac{2\tan A}{1-\tan^2 A}$ *	M1 A1cso* [2]
(b)	$\tan A-\dfrac{2\tan A}{1-\tan^2 A}=0$ $\tan A(1-\tan^2 A)-2\tan A=0$ $\tan A-\tan^3 A-2\tan A=0$ $\tan^3 A+\tan A=0$ $\tan A(\tan^2 A+1)=0$ $\tan A=0 \qquad A=0, 180$ $\left(\tan^2 A=-1 \qquad \text{no solutions}\right)$	M1 M1 M1 A1, A1 [5]
ALT	$\tan A-\dfrac{2\tan A}{1-\tan^2 A}=0$ $\tan A\left(1-\dfrac{2}{1-\tan^2 A}\right)=0 \Rightarrow 1-\tan^2 A-2=0 \text{ or } 2=1-\tan^2 A$ $\tan^2 A=-1$ $\tan A=0 \qquad A=0, 180$ $\left(\tan^2 A=-1 \qquad \text{no solutions}\right)$	
(ii)	$\cos x\cos\dfrac{\pi}{6}+\sin x\sin\dfrac{\pi}{6}\left(=\sin x\right)$ $\sin x+\sqrt{3}\cos x=2\sin x \Rightarrow \sqrt{3}=\tan x$ $x=-\dfrac{2\pi}{3},\dfrac{\pi}{3},\dfrac{4\pi}{3}$	M1 dM1 A1 A1 [4]
		Total 11 marks

Type MS · Turning sin and cos equations into tan

Answer 8 of 9 · Question 7 part b.ii · 4 marks

2022 JUN · Paper 02 · extract 1 of 3 · find part b.ii in this mark scheme extract

7(i) (a)	$\tan 2\theta = \dfrac{\tan\theta + \tan\theta}{1-\tan\theta\tan\theta}$	M1
	$= \dfrac{2\tan\theta}{1-\tan^2\theta}$ *	A1 cso (2)
(b)	$1 = \dfrac{2\tan\alpha}{1-\tan^2\alpha}$	
	$\tan^2\alpha + 2\tan\alpha - 1 = 0$	M1
	$\tan\alpha = -1 \pm \sqrt{2}$	M1 A1 (3)
(ii) (a)	$\cos(x-30) = \sin(x+30)$	
	$\Rightarrow \cos x\cos 30^\circ + \sin x\sin 30^\circ = \sin x\cos 30^\circ + \cos x\sin 30^\circ$	M1
	$\left(\dfrac{\sqrt{3}}{2} - \dfrac{1}{2}\right)\cos x = \left(\dfrac{\sqrt{3}}{2} - \dfrac{1}{2}\right)\sin x$	M1
	$\dfrac{\sin x}{\cos x} = \tan x = \dfrac{\left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right)}{\left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right)}$	M1
	$\tan x = 1$ *	A1 cso (4)
(b)	$\tan 2y = 1 \Rightarrow \tan y = -1 \pm \sqrt{2}$	
	$y = -67.5^\circ, 22.5^\circ$	M1 A1 (2)
		Total 11 marks

Part	Mark	Notes
(i)(a)	**M1**	For use of $\tan(A+B) \Rightarrow \tan 2\theta = \dfrac{\tan\theta+\tan\theta}{1-\tan\theta\tan\theta}$ and simplifying to obtain the given result.
	A1	For the given result with no errors seen
(i)(b)	**M1**	For setting $1 = \dfrac{2\tan\alpha}{1-\tan^2\alpha}$ and for rearranging to obtain a 3TQ with terms in any order
	M1	For a correct attempt using a valid method to solve the 3TQ with two solutions for $\tan\alpha$ Accept use of θ for example in place of α
	A1	For $\tan\alpha = -1 \pm \sqrt{2}$
(ii)(a)	**General principle**. (The second and third M marks can be carried out in either order) First M1 – For expanding the two addition formulae **correctly and** equating. Second M1 – For collecting up like terms to obtain $\sin x(.....) = \cos x(.....)$ allow sign errors. Third M1 – For use of the $\tan = \dfrac{\sin}{\cos}$ identity Final A1 – For the given answer with no errors	

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Answer 8 of 9 · Question 7 part b.ii · 4 marks

2022 JUN · Paper 02 · extract 2 of 3 · find part b.ii in this mark scheme extract

	They can work in terms of sin30°, cos30° or tan30° or alternatively $\frac{1}{2}, \frac{\sqrt{3}}{2}, \frac{1}{\sqrt{3}}$ respectively	
	M1	For the correct use of $\cos(A-B)$ and $\sin(A+B)$ to obtain $\cos x\cos 30^\circ + \sin x\sin 30^\circ = \sin x\cos 30^\circ + \cos x\sin 30^\circ$
	M1	For collecting up like terms to obtain: $\sin x(\sin 30^\circ - \cos 30^\circ) = \cos x(\sin 30^\circ - \cos 30^\circ)$ **NB**: This is an A mark in Epen
	M1	For use of $\tan = \frac{\sin}{\cos}$ at any stage in their working.
	A1	For $\tan x = 1$
	ALT	
	M1	For the correct use of $\cos(A-B)$ and $\sin(A+B)$ to obtain $\cos x\cos 30^\circ + \sin x\sin 30^\circ = \sin x\cos 30^\circ + \cos x\sin 30^\circ$
	M1	For dividing through by $\cos x$ and collecting up like terms to obtain: $\frac{\cos x\cos 30^\circ + \sin x\sin 30^\circ}{\cos x} = \frac{\sin x\cos 30^\circ + \cos x\sin 30^\circ}{\cos x}$ $\Rightarrow \frac{\sin x}{\cos x}(\sin 30^\circ - \cos 30^\circ) = \sin 30^\circ - \cos 30^\circ$ **OR** For dividing through by $\cos x$ **and** $\cos 30^\circ$ and also collecting up like terms to obtain: $\frac{\cos x\cos 30^\circ + \sin x\sin 30^\circ}{\cos x\cos 30^\circ} = \frac{\sin x\cos 30^\circ + \cos x\sin 30^\circ}{\cos x\cos 30^\circ}$ $\Rightarrow 1 + \frac{\sin x\sin 30^\circ}{\cos x\cos 30^\circ} = \frac{\sin x}{\cos x} + \frac{\sin 30^\circ}{\cos 30^\circ} \Rightarrow \frac{\sin x}{\cos x}\left(\frac{\sin 30^\circ}{\cos 30^\circ} - 1\right) = \frac{\sin 30^\circ}{\cos 30^\circ} - 1$
	M1	For use of $\tan = \frac{\sin}{\cos}$ at any stage in their working.
	A1	For $\tan x = 1$
(ii)(b)	**M1**	For solving the equation $\tan 2y = 1 \Rightarrow \tan y = -1 \pm \sqrt{2} \Rightarrow y = ..., ...$ (leading to two values for y) **NB**: This is an B mark in Epen
	A1	$y = -67.5^\circ, 22.5^\circ$ Both values required. **NB**: This is an B mark in Epen
	ALT	
	M1	For solving the equation $\tan 2y = 1$ to achieve at least one correct value for $2y$ $2y = \tan^{-1}1 = 45^\circ, -135^\circ$ [Accept even for example 225° for this mark]
	A1	$y = -67.5^\circ, 22.5^\circ$ Both values required. For extra angles in range withhold this mark. Extra angles outside of the range – ignore. **NB**: This is an B mark in Epen

Type MS · Turning sin and cos equations into tan

Answer 8 of 9 · Question 7 part b.ii · 4 marks

2022 JUN · Paper 02 · extract 3 of 3 · find part b.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Turning sin and cos equations into tan

Answer 9 of 9 · Question 11 part a · 4 marks

2020 JAN · Paper 02 · extract 1 of 4 · find part a in this mark scheme extract

11(a)	$(3\sin A\cos B - 3\cos A\sin B) = (\sin A\cos B + \cos A\sin B)$	M1
	$2\sin A\cos B = 4\cos A\sin B$	M1
	$\rightarrow \frac{\sin A}{\cos A} = 2\frac{\sin B}{\cos B}$	M1
	$\tan A = 2\tan B \quad k = 2$	A1 (4)
(b)	$\frac{(\cos^4\theta - \sin^4\theta)}{\cos^2\theta} = \frac{(\cos^2\theta + \sin^2\theta)(\cos^2\theta - \sin^2\theta)}{\cos^2\theta}$	M1
	$= \frac{(\cos^2\theta - \sin^2\theta)}{\cos^2\theta}$	M1
	$= 1 - \tan^2\theta$ *	A1 cso (3)
ALT 1	$1-\tan^2\theta = 1-\frac{\sin^2\theta}{\cos^2\theta} = \frac{\cos^2\theta-\sin^2\theta}{\cos^2\theta}$	M1
	$= \frac{\cos^2\theta-\sin^2\theta}{\cos^2\theta}\times\left(\cos^2\theta+\sin^2\theta\right)$	M1
	$= \frac{\cos^4\theta-\sin^4\theta}{\cos^2\theta}$	A1
ALT 2	$\frac{\cos^4\theta-\sin^4\theta}{\cos^2\theta} = \cos^2\theta - \frac{\sin^4\theta}{\cos^2\theta} = \cos^2\theta - \tan^2\theta\sin^2\theta$	M1 Eliminate 4th powers
	$= \cos^2\theta - \tan^2\theta\left(1-\cos^2\theta\right)$	M1 Eliminate $\sin^2$
	$= \cos^2\theta - \tan^2\theta + \sin^2\theta = 1-\tan^2\theta$	A1
(c)(i)	$\cos(45-30)$ or $\cos(60-45) = \frac{1}{2}\times\frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2}\times\frac{\sqrt{2}}{2}$	M1
	$= \frac{\sqrt{2}}{4} + \frac{\sqrt{6}}{4} = \frac{\sqrt{2}+\sqrt{6}}{4}$ *	A1cso (2)
ALT	By using double angle formula: $\cos^2 15° = \frac{1}{2}\left(1+\cos 30°\right) = \frac{1}{2}\left(1+\frac{\sqrt{3}}{2}\right)$	M1
	Leading to the *given* answer. $\cos 15° = \sqrt{\left(\frac{2+\sqrt{3}}{4}\right)}$ or $\frac{\sqrt{2+\sqrt{3}}}{2}$ must be seen.	A1

Type MS · Turning sin and cos equations into tan

Answer 9 of 9 · Question 11 part a · 4 marks

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Question Number	Scheme	Marks
(ii)	$\tan 255 = \tan 75$	B1
	$= \tan(30+45) = \dfrac{\tan 30 + \tan 45}{1 - \tan 30 \tan 45}$	M1
	$= \dfrac{\frac{\sqrt{3}}{3}+1}{1-\frac{\sqrt{3}}{3}}$	dM1
	$\dfrac{\frac{3+\sqrt{3}}{3}}{\frac{3-\sqrt{3}}{3}} = \dfrac{3+\sqrt{3}}{3-\sqrt{3}}$ *	A1cso (4)
		[13]
(a)		
M1	Expand both sides of the equation using correct formulae	
M1	Collect like terms from their expansions. (Not dependent)	
M1	Divide through by $\cos A \cos B$	
A1	Replace each fraction with the appropriate tangent and show $k = 2$ (value need not be shown explicitly)	
(b)		
M1	Factorise the numerator using the difference of 2 squares.	
M1	Replace $\sin^2\theta + \cos^2\theta$ with 1	
A1cso	Divide both terms by $\cos^2\theta$ and obtain the *given* answer with no errors seen.	
ALT 1		
M1	Use $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$ and obtain a single fraction with no tan	
M1	Indicate multiplication by $\sin^2\theta + \cos^2\theta$	
A1cso	Multiply and obtain the *given* answer with no errors seen.	
(c)(i)		
M1	Express 15 as the difference of 2 suitable numbers, expand using a correct formula and substitute the correct exact values for the trig functions (substitution must be shown).	
A1cso	Simplify and combine the fractions to obtain the *given* answer with no errors seen.	
(ii)		
B1	$\tan 255 = \tan 75$ seen explicitly or used. OR eg tan(210 + 45) – give B1 for tan 210 = tan 30 used	
M1	Express 75 as 30 + 45 and expand $\tan(30+45)$ using the correct formula (given on the formula page) OR expand tan (210 + 45) If 75 = 15 + 60 is used tan 15 can be obtained from a calculator but must be in exact form..	
dM1	Substitute the correct exact values for the trig functions	
A1cso	Simplify the fractions to obtain the *given* answer with full working and no errors seen.	

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